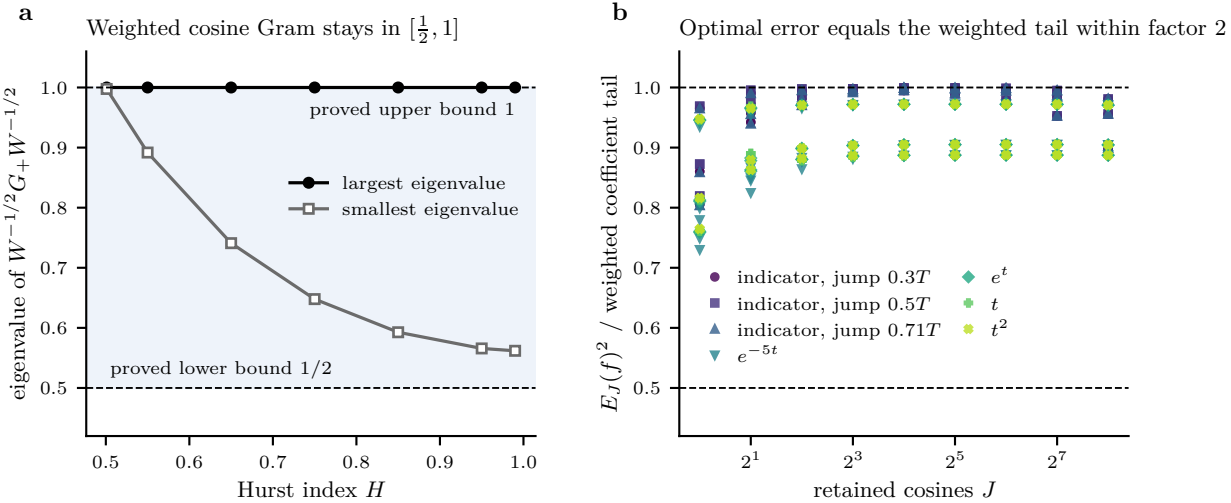

a
Weighted cosine Gram stays in $[\frac{1}{2}, 1]$
eigenvalue of $W^{-1/2} G_+ W^{-1/2}$
1.0
0.9
0.8
0.7
0.6
0.5
proved upper bound 1
largest eigenvalue
smallest eigenvalue
proved lower bound 1/2
0.5
0.6
0.7
0.8
0.9
1.0
Hurst index H
b
Optimal error equals the weighted tail within factor 2
$E_J(f)^2$ / weighted coefficient tail
1.0
0.9
0.8
0.7
0.6
0.5
indicator, jump $0.3T$
indicator, jump $0.5T$
indicator, jump $0.71T$
e^{-5t}
e^t
t
t^2
2^1
2^3
2^5
2^7
retained cosines J